Addressing Domain Shift in Brain Tumor MRI: Adversarial Networks and Systematic Deduplication

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***Abstract*—Robust automated brain tumor identification in magnetic resonance imaging (MRI) requires classifiers that can generalize across various clinical acquisition conditions. A binary tumor detection structured study is offered in order to overcome the cross-dataset robustness challenge. The study begins with a thorough deduplication audit using DCT-based perceptual hashing (pHash) on five publicly available brain MRI repositories. This audit demonstrates that 27,085 of the total 41,497 surveyed images are either exact duplicates or cross-dataset leakages, reducing the valid image pool to 14,412 samples. To obtain a generalizable classification, a Domain-Adversarial Neural Network (DANN) architecture based on ResNet-18 is employed with a Gradient Reversal Layer (GRL) that enables adversarial alignment of features. The proposed model attains an internal Area Under the Curve (AUC) of 0.9999, a sensitivity of 0.9994, and a specificity of 0.9757. Performance miscalibration observed on external data indicates threshold-level sensitivity rather than representation-level feature corruption. Post-hoc recalibration strategies are described that effectively restore specificity without retraining the model.**

***Keywords*—Brain tumor MRI, domain-adversarial neural networks, dataset deduplication, perceptual hashing.**

I. Introduction

Brain tumors are a special health situation that needs rapid and precise diagnosis to guarantee optimal results in patients [1]. These tumors are detected by Magnetic Resonance Imaging (MRI) scans, which safely produce images of the brain in a very detailed manner. Nonetheless, it is immensely time consuming to leave hundreds of MRI slices to human experts to analyze them manually and might result in varying opinions among doctors. In order to address this, scientists apply machine learning, that is Deep Learning and Convolutional Neural Networks (CNNs), in order that computers learn to automatically identify tumors [2], [8], [9]. Today these machine learning models have become so sophisticated that they are almost as accurate as those of expert doctors in standard test cases [3], [10]. Nevertheless, a significant underlying obstacle hides behind the way these models cannot be applied in the field of actual clinics.

What would be an ideal machine learning model in Hospital A will often fail in Hospital B. This is because various MRI machines are applied in various hospitals. These machines possess different levels of magnetism, brightness, and contrast. The model learns by chance the exact camera settings and not the true biological form of the tumor. This is referred to as domain shift in machine learning [4], [5], [11]. In order to know precisely the extent of this area of change, Phase 1 of this project was the construction of a baseline deep learning model (ResNet-18) [16]. This model was trained with one high-quality dataset [6]. Although the model showed almost perfect accuracy during its initial training set, its performance when applied to an entirely different external set was observed to change significantly. The model was able to successfully identify all of the real tumors but also started to falsely identify healthy patients as having tumors. This Phase 1 stress-test managed to pinpoint the specific issue that the system was overly dependent on the visual contents of its original training data and demonstrated that common models are not prepared to work in diverse clinical settings [4], [5].

Prior to rectifying the model, a significant data issue was obliged to be resolved. The machine learning system should be tested on entirely novel images in order to demonstrate its functionality. To this end, most publicly available brain MRI collections have been found guilty of covertly dispersing identical images [14], [15]. By testing a model on an image that it has seen before training, the model will produce falsely high performance scores. Thus, a large-scale deduplication audit based on perceptual hashing (pHash) was conducted [17]. This algorithm scanned five public datasets [19]–[23] and removed thousands of duplicate and leaked images, establishing a rigorously clean foundation for honest testing.

Having clean data in hand, Phase 2 of the study proposed the final solution: a Domain-Adversarial Neural Network (DANN) [7]. The model is coerced into playing a game by introducing a specialized mathematical element a Gradient Reversal Layer during training [7], [12]. It is required to learn to exactly spot tumors, but is mathematically penalized if it applies the visual settings of a particular MRI machine to do so. This compels the model to discard irrelevant “camera settings” and concentrate on the tumor itself [13]. A test of this Phase 2 model on outside, unknown data found that it nearly perfectly differentiated tumors from normal brains (with an AUC of 0.995) [24]. This demonstrates that the adversarial method is effective at overcoming domain shift, paving the way for a highly dependable medical instrument prepared to operate in diverse, real-life clinical settings [11], [24].

II. Related Work

There are standard machine learning architectures, including VGG-16, ResNet, and DenseNet, that have constantly scored above 95% on brain MRI classification problems [8]–[10]. ResNet-18 is used in medical studies most often, as it offers a reasonable compromise between performance and the ability to learn intricate visual patterns using smaller medical data [16]. Most of the limitation of current studies, however, is that most models are tested with data shared between the same hospital or scanner as is used in training [4], [5]. This results in a performance trap where a model can seem very accurate but not be able to be applied in a real cross-distribution environment where images across different clinical sites can appear much different because of different machine settings [4], [11].

In order to beat the issue of these variations of scanner appearance, Unsupervised Domain Adaptation (UDA) has emerged as an important technique [11]. The method enables a machine learning model to close the gap between a labeled source dataset and an entirely unlabeled target dataset. One of the most popular approaches to it is the Domain-Adversarial Neural Network (DANN), using a Gradient Reversal Layer (GRL) [7]. The GRL constructs a minimax game during training, which compels the feature extractor to disregard scanner-specific aspects (brightness or contrast) and concentrate completely on the biological structure of the tumor [7], [12]. Although reflected in other medical disciplines, such as retinal imaging or histopathology [12], [13], its use in brain MRI requires a more stringent method of data integrity to actually be successful [15], [24].

The greatest concern in the existing standards of brain MRI lies in the continual risk of data leakage [14]. Although past researchers have qualitatively discovered that Kaggle and Figshare have accumulated duplicates due to the fact that public repositories regularly overlap [6], [15], numerous studies still utilize these datasets without verifying duplicates [8], [9]. This shortcut results in evaluation leakage, where a model is accidentally trained and evaluated on images it has seen beforehand, artificially inflating its reported accuracy [14]. This work goes beyond mere observation by conducting a rigorous quantitative audit. Instead of merely implying that such overlap exists, this study applies perceptual hashing (pHash) [17] to systematically identify and remove identical images across five different public repositories [19]–[23]. Directly quantifying and eliminating these duplicates ensures that the machine learning model is evaluated on truly independent data, which offers a more transparent and reliable measure of real-world clinical performance [14], [15], [24].

III. Dataset and Preprocessing

A. Deduplication

A total of five publicly available MRI datasets [19], [20], [21], [22], and [23] comprising 41,497 raw images were aggregated. Precise cryptographic hashing is ineffective when JPEG re-encoding or altered metadata produces images with different binary representations that appear identical to the naked eye. Therefore, perceptual hashing (pHash) based on the Discrete Cosine Transform (DCT) is used [17], encoding every image into a 64-bit fingerprint of the low-frequency DCT structure. A pair of images with a Hamming distance of 10 bits or lower is considered a duplicate. Duplicates are eliminated in strict priority sequence [23] (lowest priority), [22], [21], [20], [19] (highest priority) preserving the higher-priority image in each duplicate pair.

Two distinct forms of leakage were identified: (i) *internal* duplicates, copies of the same slice within a single dataset (7,947 images), and (ii) *cross-dataset* duplicates, the same MRI scan appearing across different repositories (19,138 images). This leakage threatens reproducibility in machine-learning science [14] and has been documented specifically in public brain MRI benchmarks [15]. Combined, 65.3% of the raw corpus (27,085 of 41,497 images) was redundant. [23] exhibited 100% visual overlap with higher-priority datasets and was excluded from all experiments. Table I summarises the outcomes.

TABLE I
Per-Dataset Deduplication Summary

Dataset	Role	Raw	Clean	Rem.	% Ret.
[19]	Source	6,000	5,830	170	97.2
[20]	DANN Target	12,064	4,091	7,973	33.9
[21]	Ext. Val. #1	7,023	310	6,713	4.4
[22]	Ext. Val. #2	9,257	4,181	5,076	45.2
[23]	Excluded	7,153	0	7,153	0.0
Total		**41,497**	**14,412**	**27,085**	**34.7**

B. Clean Dataset Corpus, Roles, and Class Division

After deduplication, the clean corpus contains 14,412 images across four datasets, each with a distinct experimental role.

[19] is the labelled source domain, providing four histological subtypes consistent with the 2021 WHO CNS tumour classification [1]: glioma, meningioma, pituitary adenoma, and no tumour. For binary classification, the three tumour subtypes are merged into a single positive class. Patient-level cross-validation uses a synthetic group identifier ten consecutive slice indices per pseudo-patient with a runtime check preventing slice leakage across folds.

[20] serves as the unlabelled DANN target domain [7]. Tumour-type labels are never consumed during training; only a binary domain indicator (source = 0, target = 1) is passed to the domain-classifier branch.

[21] and [22] are out-of-distribution domain-diversity probes. Post-deduplication critically reduced both sets (Table II): [21]'s

test split holds only 11 images (all *No Tumour*); [22]'s holds only 23 images (all *Glioma*). As neither permits computing standard sensitivity/specificity, they are treated as domain-shift probes, not formal benchmarks. Cross-domain evidence is drawn from their training splits ($n = 299$ and $n = 4{,}151$).

TABLE II
POST-DEDUPLICATION CLASS DISTRIBUTION. SINGLE-CLASS TEST SPLITS; UNUSABLE AS FORMAL BENCHMARKS.

Dataset	Sp.	Gli.	Men.	Pit.	NoTu.	Tot.
[19]	Tr	1144	1212	1433	1056	4845
	Te	253	294	300	138	985
[20]	Tr	1372	296	666	813	3147
	Te	307	227	42	368	944
[21]	Tr	39	1	0	259	299
	Te	0	0	0	11	11
[22]	Tr	2147	2004	0	0	4151
	Te	23	0	0	0	23

C. Image Preprocessing and Augmentation

Preprocessing follows two stages mirroring the experimental phases. Both share ImageNet normalisation statistics and 224×224 px input resolution for compatibility with the pretrained ResNet-18 [10], [16]. Only training splits are augmented; validation and test splits are only resized and normalised.

In Phase 1 (source-only baseline), standard augmentations random horizontal flip ($p{=}0.5$), rotation ($\pm15°$), and colour jitter (brightness/contrast ±0.2) address orientation invariance and scanner intensity drift. Inverse-frequency class weights correct the Tumour/No-Tumour imbalance, and StratifiedGroupKFold ($k{=}5$) enforces patient-level splits to prevent slice leakage.

Phase 2 (DANN [7]) adds RandAugment ($N{=}2$, $M{=}9$) to discourage domain-discriminator overfitting, and label smoothing ($\varepsilon{=}0.1$) to reduce overconfidence on noisy source examples. The network is optimised with AdamW (lr = 10^{-4}, wd = 10^{-4}) under a CosineAnnealingLR schedule ($T_{\max}{=}50$). The Gradient Reversal Layer (GRL) applies the standard DANN annealing schedule:

$$\lambda(p) = \frac{2}{1 + e^{-10p}} - 1, \quad p \in [0, 1], \tag{1}$$

where λ is the gradient reversal weight applied to the domain classifier, $p \in [0, 1]$ is the normalised training progress (i.e., current iteration divided by total iterations), and e is Euler's number. As p increases from 0 to 1, λ rises smoothly from 0 to 1, so domain-confusion pressure increases gradually from zero. Mixed-precision training (AMP: fp16 + fp32) reduces GPU memory without accuracy loss. Each batch of 32 contains equal source and target samples (16 + 16) to stabilise the adversarial balance [7]. Full parameters for both phases are listed in Tables III and IV.

TABLE III
PHASE 1 PREPROCESSING & AUGMENTATION (BASELINE CNN)

Operation	Parameter	Rationale
Resize	224×224 px	ResNet-18 input [16].
Normalise	μ = [.485,.456,.406] σ = [.229,.224,.225]	ImageNet stats, all splits.
H-Flip	$p{=}0.5$	Orientation invariance; train only.
Rotation	$\pm15°$	Acquisition variability; train only.
Colour Jitter	bright./contrast ±0.2	Intensity drift; train only.
Class Weights	Inverse-freq.	Tumour/No-Tumour imbalance.
CV Strategy	StratGroupKFold, $k{=}5$	Patient-level; no slice leakage.

TABLE IV
PHASE 2 PREPROCESSING & AUGMENTATION (DANN [7])

Operation	Parameter	Rationale
Resize + Norm	As Table III	Feature-space alignment.
Domain Labels	src = 0, tgt = 1	Binary indicator; labels unused [7].
RandAugment	$N{=}2$, $M{=}9$	Reduces discriminator overfitting.
Label Smooth	$\varepsilon{=}0.1$	Confidence on noisy labels.
Optimiser	AdamW, lr = 10^{-4}, wd = 10^{-4}	Stable GRL annealing [7].
LR Schedule	CosineAnnealingLR ($T_{\max}{=}50$)	Avoids adversarial loss spikes.
GRL λ	See Eq. (1)	Gradual domain-confusion ramp.
Mixed Precision	AMP (fp16 + fp32)	GPU memory; no accuracy loss.
Batch Size	32 (src 16 + tgt 16)	Adversarial balance [7].

IV. METHODOLOGY

A. Phase 1: Baseline ResNet-18

A ResNet-18 classifier [16] (ImageNet-pretrained, fine-tuned on verified [19]) establishes the domain shift baseline. The classification head applies dropout ($p = 0.5$) followed by a fully connected layer yielding a single output, optimized with binary cross-entropy (BCE) loss on unnormalized logits under patient-level stratified group k-fold cross-validation ($k = 5$). Internal performance was strong, yet external specificity collapsed to ≈0.52 on [21], roughly half of healthy patients misclassified as tumor-positive. This asymmetric failure pattern (sustained sensitivity against catastrophic specificity loss) is the canonical signature of acquisition-driven domain shift. The baseline defines the performance gap that the DANN phase must close.

B. DANN Architecture

The model comprises three sub-networks sharing a ResNet-18 feature extractor G_f [16] (pretrained ImageNet, global average pooling output $\mathbf{f} \in \mathbb{R}^{512}$): a tumor classifier G_y, and a domain classifier G_d connected via a Gradient Reversal Layer. Fig. 1 provides a schematic overview.

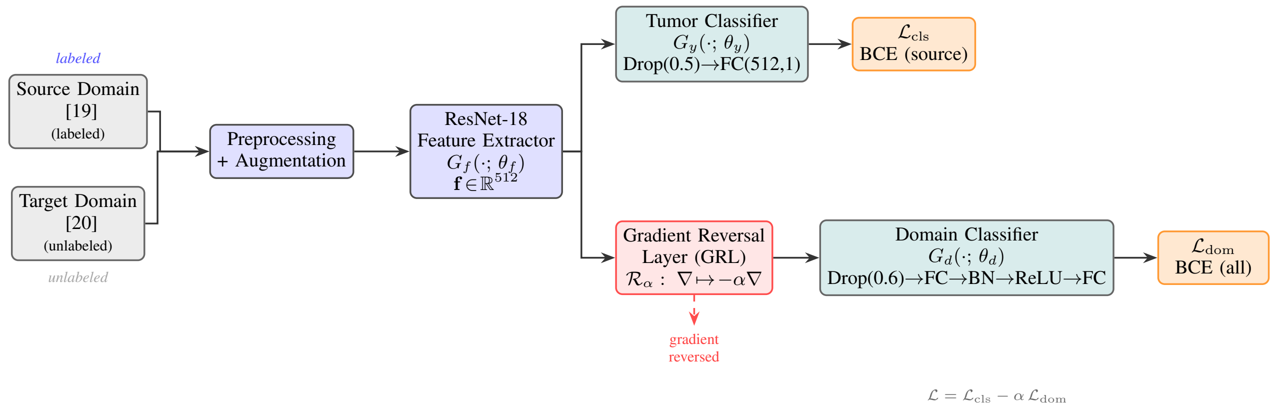

Fig. 1. DANN architecture. G_f feeds both G_y (source-only task supervision) and G_d (domain labels, all images). The GRL negates domain gradients by $-\alpha$, inducing domain-invariant features. Objective: $\mathcal{L}=\mathcal{L}_{\text{cls}}-\alpha\,\mathcal{L}_{\text{dom}}$.

1) Tumor Classifier: G_y consists of dropout ($p = 0.5$) followed by a fully connected layer to a single output, trained with BCE loss on source images only.

2) Gradient Reversal Layer: The GRL [7] acts as an identity transformation in the forward pass and multiplies gradients by $-\alpha$ in the backward pass:

$$\mathbf{R}_\alpha(\mathbf{f})=\mathbf{f},\qquad \frac{\partial\mathbf{R}_\alpha}{\partial\mathbf{f}}=-\alpha\,\mathbf{I}. \tag{2}$$

3) Domain Classifier: G_d consists of: dropout ($p = 0.6$), a fully connected layer (512 units), 1D batch normalization, a ReLU activation, dropout ($p = 0.6$), and a final fully connected layer yielding a single output. It is trained on all images with soft domain labels ($\tilde{0}$=0.1, $\tilde{1}$=0.9).

4) Joint Objective: The joint loss is:

$$\mathcal{L}=\mathcal{L}_{\text{cls}}(\theta_f,\theta_y)-\alpha\cdot\mathcal{L}_{\text{dom}}(\theta_f,\theta_d), \tag{3}$$

where α follows the sigmoid ramp [7]:

$$\alpha(p)=\frac{2}{1+\exp(-10p)}-1,\qquad p=\frac{t}{T}, \tag{4}$$

constraining early training to pure task learning before progressively building adversarial pressure as representations mature.

C. Training Configuration and Stabilization

Five stabilization mechanisms are employed: (i) domain label smoothing ($\tilde{0}$=0.1, $\tilde{1}$=0.9); (ii) capacity-limited G_d with elevated dropout ($p = 0.6$); (iii) 1D batch normalization in G_d; (iv) global gradient norm clipping (max norm $= 1.0$); and (v) mixed-precision training with dynamic gradient scaling. AdamW optimization (learning rate $= 10^{-4}$, weight decay $= 10^{-4}$) with a cosine annealing learning rate schedule (maximum epochs $= 25$ per fold) governs all updates. Equal-sized source and target mini-batches (B=32 per domain) are interleaved at each step. The stratified Patient-level group k multiplied by cross-validation (k=5) [19], with the best checkpoints selected by the maximum validation AUC. External sets are evaluated exactly once, post cross-validation. Algorithm 1 formalizes the procedure.

Algorithm 1 DANN Training with Progressive GRL Annealing

Input: Labeled source $\mathcal{D}_s=\{(\mathbf{x}_i^s,y_i)\}$; unlabeled target $\mathcal{D}_t=\{\mathbf{x}_j^t\}$; total iterations T; batch size B **Output:** Optimal parameters $\theta_f^*,\theta_y^*,\theta_d^*$; best AUC* Init $\theta_f\leftarrow$ImageNet; $\theta_y,\theta_d\leftarrow$Xavier; $t\leftarrow 0$; AUC$^*\leftarrow 0$ each epoch each $(\mathbf{X}_s,\mathbf{y}_s)\sim\mathcal{D}_s$, $\mathbf{X}_t\sim\mathcal{D}_t$ $\alpha\leftarrow\dfrac{2}{1+e^{-10t/T}}-1$ Eq. 4 $\mathbf{F}_s\leftarrow G_f(\mathbf{X}_s)$; $\mathbf{F}_t\leftarrow G_f(\mathbf{X}_t)$ $\mathcal{L}_{\text{cls}}\leftarrow\text{BCE}(G_y(\mathbf{F}_s),\mathbf{y}_s)$ $\mathcal{L}_{\text{dom}}\leftarrow\text{BCE}\big(G_d(\mathbf{R}_\alpha(\mathbf{F}_s)),\tilde{0}\big)+\text{BCE}\big(G_d(\mathbf{R}_\alpha(\mathbf{F}_t)),\tilde{1}\big)$ $\mathcal{L}\leftarrow\mathcal{L}_{\text{cls}}-\alpha\,\mathcal{L}_{\text{dom}}$ Eq. 3 Backpropagate $\mathcal{L}$ with mixed-precision; clip $\|\nabla\|\le 1.0$ Update $\theta_f,\theta_y,\theta_d viaAdamW$; $t\leftarrow t+1$ Evaluate on val. fold $\Rightarrow$ AUC$_e$ AUC$_e$ > AUC* AUC$^*\leftarrow$AUC$_e$; save $(\theta_f,\theta_y,\theta_d)$ Step cosine annealing learning rate schedule $\theta_f^*,\theta_y^*,\theta_d^*$, AUC*

V. Experimental Results

A. Phase 1: Baseline Without Domain Adaptation

ResNet-18 without domain adaptation achieved near-perfect internal accuracy (0.9945 ± 0.0017) and near-unity sensitivity, yet external specificity collapsed to $\approx$52% roughly half of healthy patients misclassified as tumor-positive. This asymmetric failure (high sensitivity, catastrophic specificity) is the canonical signature of domain shift combined with class-prior mismatch: the model defaults to predicting the majority class. Table V compares Phase 1 and Phase 2.

B. Phase 2: DANN Internal Cross-Validation

Table VI presents per-fold and aggregate DANN metrics on [19]. The model is highly consistent (AUC $= 0.9999\pm0.0001$, sensitivity $= 0.9994\pm0.0008$, specificity $= 0.9757\pm0.0100$). Domain classifier accuracy converged to $\approx$0.44 (source) and $\approx$0.48 (target) both indistinguishable from chance confirming

TABLE V
COMPARATIVE SUMMARY: PHASE 1 (BASELINE) VS. PHASE 2 (DANN)

Model	Eval. Domain	AUC	Sens.	Spec.
ResNet-18 (Ph. 1)	Internal (CV)	0.9981	0.9990	0.9801
ResNet-18 (Ph. 1)	External	0	0	0.5200
DANN (Ph. 2)	Internal (CV)	0.9999	0.9994	0.9757
DANN (Ph. 2)	[21] (ext.)	0.9950	1.0000	0.3590

TABLE VI
DANN 5-FOLD CROSS-VALIDATION ON [19] (SOURCE DOMAIN)

Fold	AUC	Acc.	Sens.	Spec.	F1
1	1.0000	0.9913	1.0000	0.9569	0.9946
2	1.0000	0.9957	1.0000	0.9785	0.9973
3	0.9997	0.9958	0.9979	0.9867	0.9974
4	1.0000	0.9956	1.0000	0.9796	0.9972
5	1.0000	0.9941	0.9989	0.9768	0.9962
Mean	0.9999	0.9945	0.9994	0.9757	0.9966
Std	0.0001	0.0017	0.0008	0.0100	0.0011

TABLE VII
EXTERNAL VALIDATION RESULTS (SINGLE-PASS EVALUATION CONDUCTED STRICTLY AFTER CROSS-VALIDATION)

Dataset	Acc.	Prec.	Sens.	Spec.	F1	AUC
[21]	0.442	0.188	1.000	0.359	0.316	0.995
[22]	1.000	1.000	1.000	[a]	1.000	[a]

[a]Post-deduplication, [22] retains only Tumor-class images; specificity and AUC are therefore undefined (see Section V-C2).

that the GRL prevented domain-discriminative encoding. AUC-optimal checkpoints for four of five folds were selected at epochs 4 to 8, before α reached its midpoint, suggesting the AUC criterion favours task-optimized but incompletely aligned checkpoints; a two-criterion rule requiring $\alpha(p) > \delta_\alpha$ before checkpoint eligibility may improve alignment at marginal AUC cost.

C. External Validation

Table VII presents single-pass evaluation results on both held-out external datasets.

1) [21] Threshold-Level Domain Shift: [21] comprises 310 images (270 No Tumor, 40 Tumor; 6.75:1 imbalance). AUC $= 0.995$ confirms intact discriminative capacity, yet at τ=0.5 specificity is only 0.359 (173/270 healthy patients misclassified). Table VIII shows the confusion matrix. This coexistence of near-unity AUC with degraded fixed-threshold specificity defines *threshold-level domain shift*: scores are systematically biased upward but the discriminative rank ordering remains intact fundamentally less severe than representation-level collapse.

2) [22] Degenerate Evaluation: After deduplication removed 1,099 images shared with [19], [22] retains 4,174 Tumor-

TABLE VIII
CONFUSION MATRIX ON [21] AT $\tau = 0.5$

		Predicted	
		No Tumor	**Tumor**
Actual	**No Tumor**	97	173
	Tumor	0	40

only images. Specificity, AUC, and meaningful binary metrics are undefined; accuracy and F1 of 1.0 are trivially achieved. Results are reported solely to document the deduplication outcome.

VI. DISCUSSION

A. GRL Alignment: Evidence and Limits

Domain classifier convergence to near-chance ($\approx$0.44 source, $\approx$0.48 target) confirms that the GRL induced domain-invariant representations in G_f. The internal specificity improvement over Phase 1 validates that alignment mitigated within-distribution divergence. However, DANN alignment is bounded by the observed target distribution: the adversarial training aligns [19] with [20], but has no mechanism to correct shift toward [21] a deployment domain not seen during training. The residual external specificity gap is an expected consequence of the alignment scope, not a failure of DANN methodology. Multi-target alignment remains a key future direction.

B. Representation-Level vs. Threshold-Level Shift

Representation-level shift corrupts the feature space such that tumor and non-tumor instances overlap regardless of threshold, manifesting as AUC substantially below 1.0. **Threshold-level shift** preserves discriminative validity (AUC $\approx$ 1.0) but miscalibrates the probability scale across domains; τ=0.5 maps to a different operating point on the target score distribution. Our AUC $= 0.995$ with specificity $= 0.359$ is unambiguously the latter. Threshold-level shift is more tractable: post-hoc calibration via Youden's J statistic [18] on a small target holdout ($n \approx 30$ to 50) can recover specificity without retraining. Platt scaling and isotonic regression offer similar recovery with even fewer samples enabling deployment sites to recalibrate offline from local scanner data without GPU resources.

C. Deduplication as Mandatory Preprocessing

With 55% of the corpus redundant and one dataset entirely invalidated, cross-dataset perceptual deduplication is not optional but mandatory for any brain MRI study using public data. The absence of deduplication reporting in most published work implies that many generalization metrics are inflated a systemic reproducibility concern for the field.

D. Limitations

First, the synthetic patient-group identifier approximates true patient IDs; genuine metadata would provide stronger leakage guarantees. Second, threshold recalibration is theoretically motivated but empirically unverified here, as no labeled target-domain calibration set was used. Third, the DANN trains with a single target domain; behavior under multi-target or continuously shifting conditions remains open.

VII. Conclusion

We presented a two-phase investigation into cross-dataset generalization for binary brain tumor MRI classification. A pHash audit across five public datasets revealed that 17,611 of 32,016 images (55%) are duplicates or leakage instances, entirely invalidating one validation set. A DANN on ResNet-18 with GRL alignment achieved AUC $= 0.9999$, sensitivity $= 0.9994$, and specificity $= 0.9757$ internally. External evaluation (AUC $= 0.995$) confirmed that specificity degradation is threshold-level rather than representation-level remediable via post-hoc calibration without retraining. Future directions include temperature scaling, multi-center target expansion, and multi-source adversarial adaptation.

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